Visual Studio LIVE! | San Diego

EXPERT SOLUTIONS FOR ENTERPRISE DEVELOPERS

W05 Integrating AI into Your Existing Applications Using Semantic Kernel & C#

Marcel de Vries

Global MD & CTO

Xebia, Microsoft Services

Level: Intermediate,

#VSLIVE

Next Generation Skills for Developers

Visual Studio LIVE!
EXPERT SOLUTIONS FOR ENTERPRISE DEVELOPERS | Las Vegas

Integrating AI into Your Existing Applications Using Semantic Kernel & C#

Marcel de Vries

Global MD & CTO Microsoft Services,
Xebia

Level: Intermediate

Session Survey

- Your feedback is very important to us
- Please take a moment to complete the session survey found in the mobile app
- Use the QR code or search for “Converge360 Events” in your app store
- Find this session on the Agenda tab
- Click “Session Evaluation”
- Thank you!

Agenda

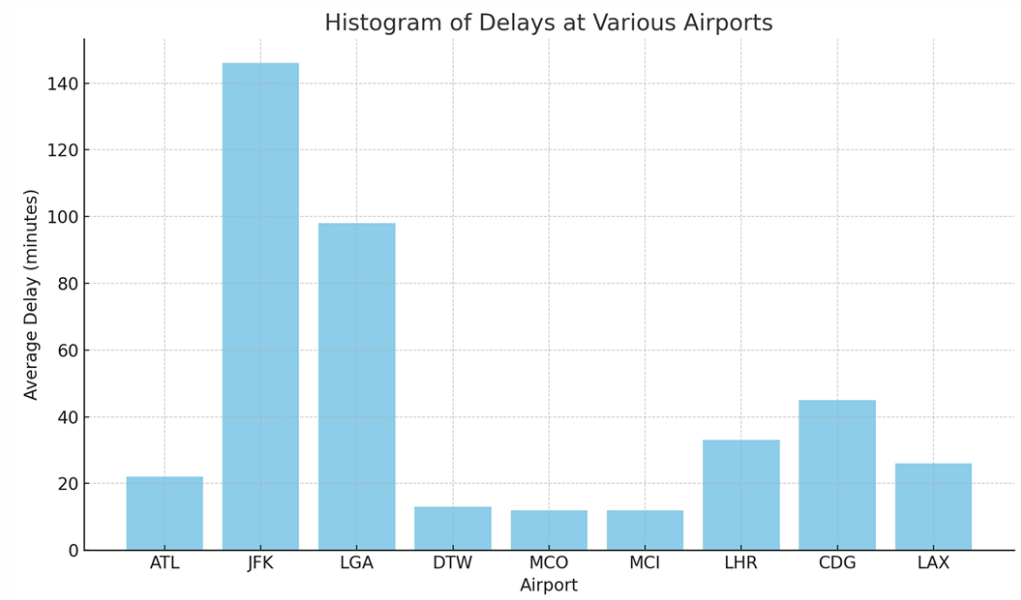
- Short intro in Large Language Models
- Introduction to Semantic Kernel
- Function calling
- Using it with your own data
- Integrating it in your application
- Summary

LLM Usage

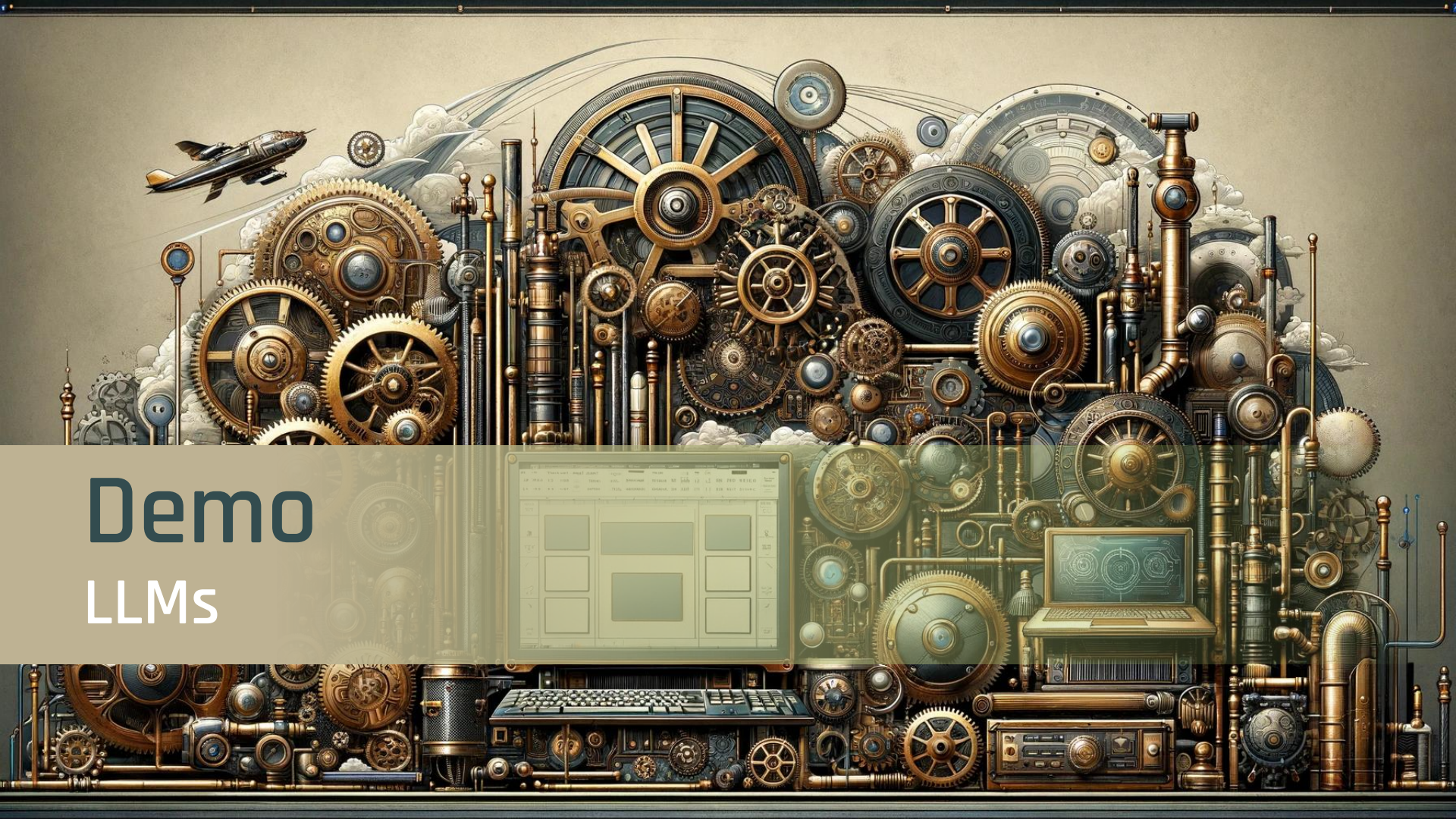
- Generating, Translating Text
- Generating, translating Code
- Generating Charts and Graphs
- Analyzing Charts

Getting data from images

Airport	Average Delay
ATL	22
JFK	146
LGA	98
DTW	13
MCO	12
MCI	12
LHR	33
CDG	45
LAX	26



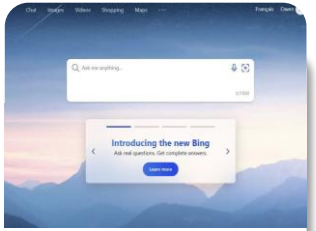
Visual Studio LIVE!
EXPERT SOLUTIONS FOR ENTERPRISE DEVELOPERS



Demo

LLMs

Modernized & augmented products with copilot

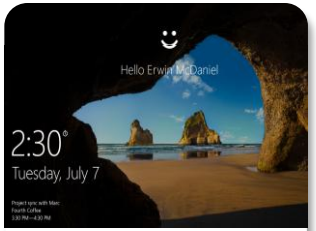


AI-based search with Microsoft Copilot

Bing

Copilot for Work across Office 365

Office

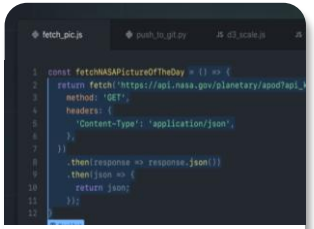


Biometric identity verification

Windows

Personalized recommendations

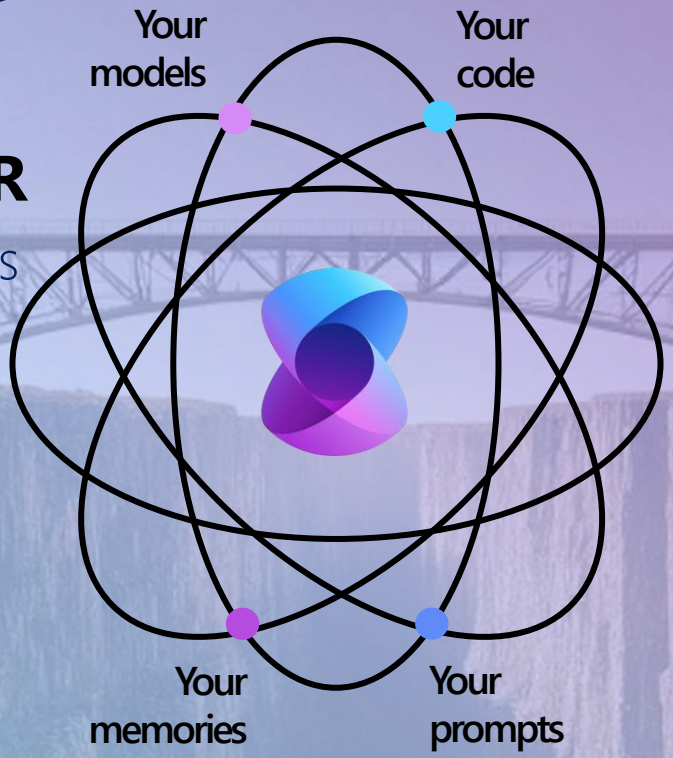
XBOX



Copilot assisted coding

GitHub

A bridge between LLMs and your code



ML RESEARCHER

keeping up with advances
cutting edge models
algorithm analysis
experimentation

APP DEVELOPER

stable and performant
efficient to deliver
easily scalable
maintainable

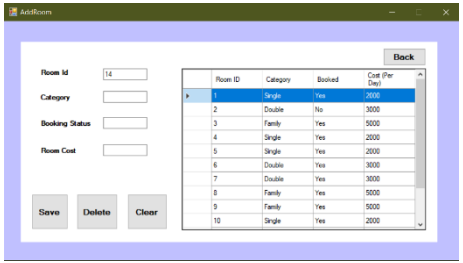
Visual Studio LIVE!
EXPERT SOLUTIONS FOR ENTERPRISE DEVELOPERS

Infuse AI in your App

Adding AI in your application can become a magical moment for your users!

Semantic Kernel

+



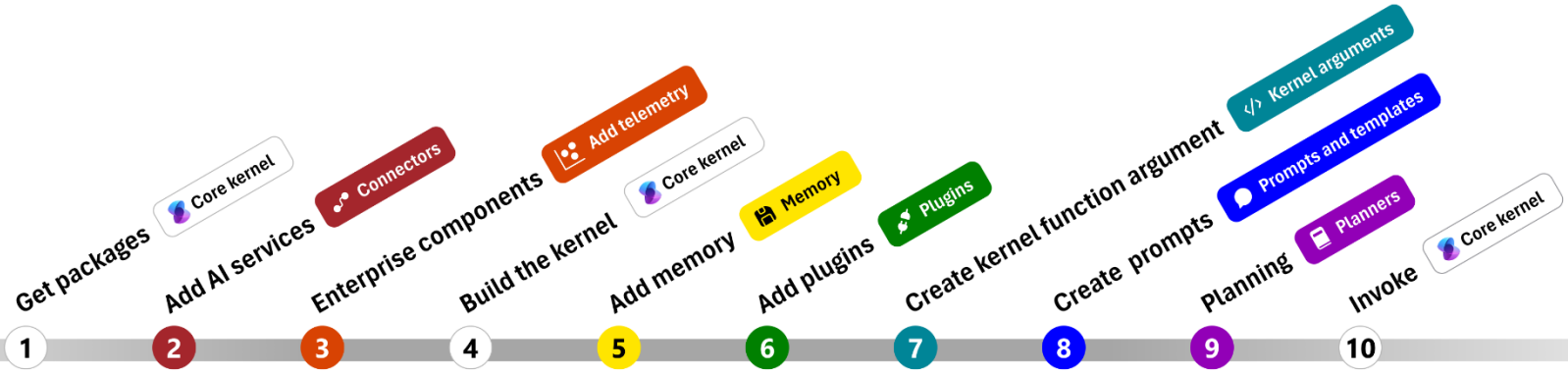
Your App

=

Innovation

Visual Studio LIVE!
EXPERT SOLUTIONS FOR ENTERPRISE DEVELOPERS

Steps using Semantic Kernel



Visual Studio LIVE!
EXPERT SOLUTIONS FOR ENTERPRISE DEVELOPERS

Getting started

```
public async Task SimplestPromptLoop(string deploymentName,
                                     string endpoint,
                                     string apiKey
{
  string response = string.Empty;

  Kernel kernel = Kernel.CreateBuilder()
        .AddAzureOpenAIChatCompletion(deploymentName, endpoint, apiKey)
        .Build();

   while (response != "quit")
   {
     Console.WriteLine("Enter your message:");
     response = Console.ReadLine();
     Console.WriteLine(await kernel.InvokePromptAsync(response));
   }
}
```

Visual Studio LIVE!
EXPERT SOLUTIONS FOR ENTERPRISE DEVELOPERS

Using IConfiguration

```
public async Task SimplestPromptLoop(string deploymentName,
                                     string endpoint, string apiKey
{
  string response = string.Empty;
  Kernel kernel = Kernel.CreateBuilder().
        AddAzureOpenAIChatCompletion(deploymentName, endpoint, apiKey).Build();
  var chatCompletionService = kernel.GetRequiredService<IChatCompletionService>();

  while (response != "quit")
   {
     Console.WriteLine("Enter your message:");
     response = Console.ReadLine();
     Console.WriteLine(await
                 chatCompletionService.GetChatMessageContentAsync(response));
   }

}
```

Visual Studio LIVE!
EXPERT SOLUTIONS FOR ENTERPRISE DEVELOPERS

Adding History

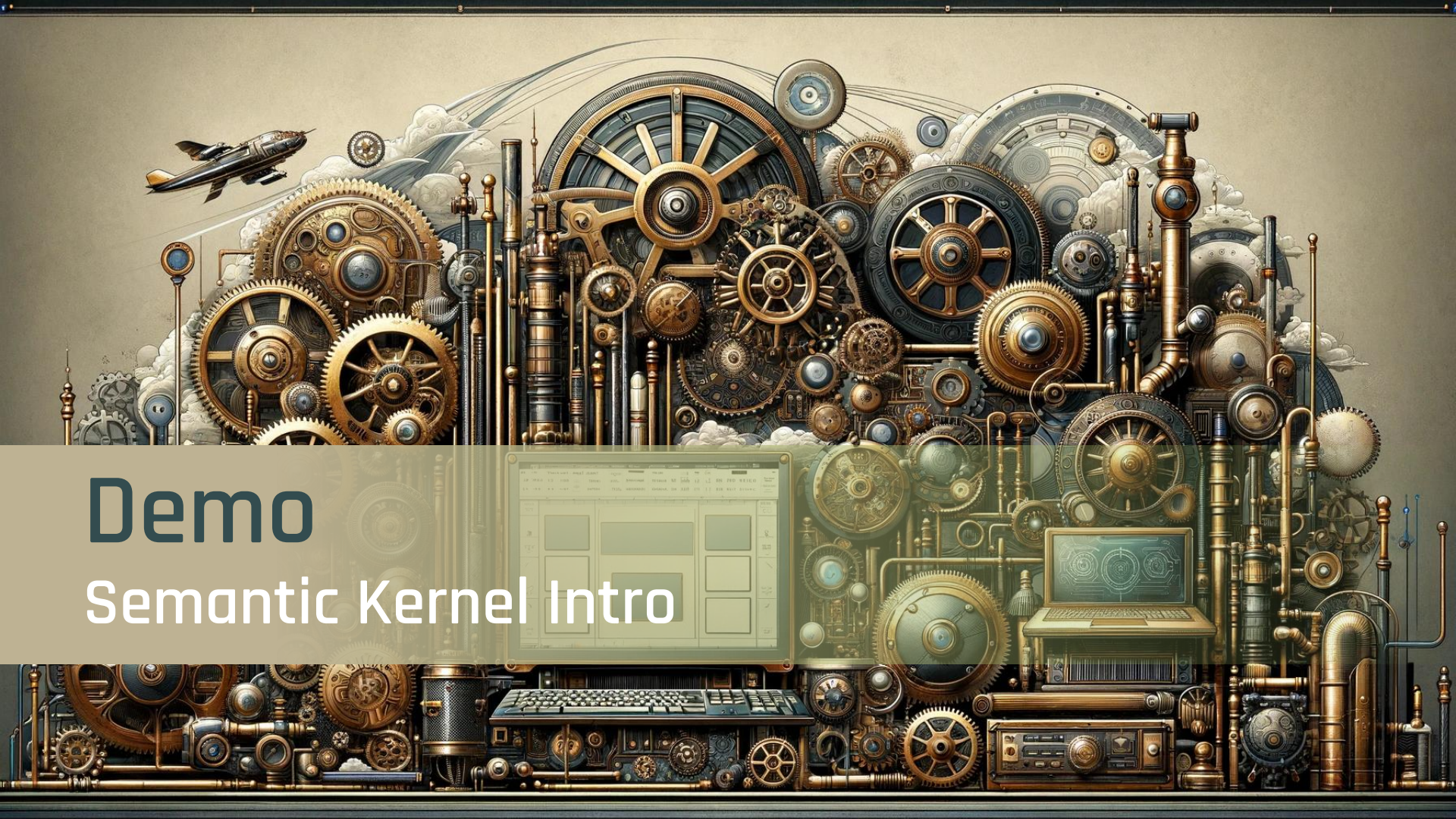
```
public async Task SimplestPromptLoop(string deploymentName,
                                     string endpoint, string apiKey
{
  string response = string.Empty;
  Kernel kernel = Kernel.CreateBuilder().
        AddAzureOpenAIChatCompletion(deploymentName, endpoint, apiKey).Build();
  var chatCompletionService = kernel.GetRequiredService<IChatCompletionService>();
  ChatHistory chatHistory = new();
  while (response != "quit")
   {
     Console.WriteLine("Enter your message:");
     response = Console.ReadLine();

        chatHistory.AddUserMessage(response);

Console.WriteLine(await
                  chatCompletionService.GetChatMessageContentAsync(chatHistory));

        chatHistory.Add(assistantMessage);
   }
}
```

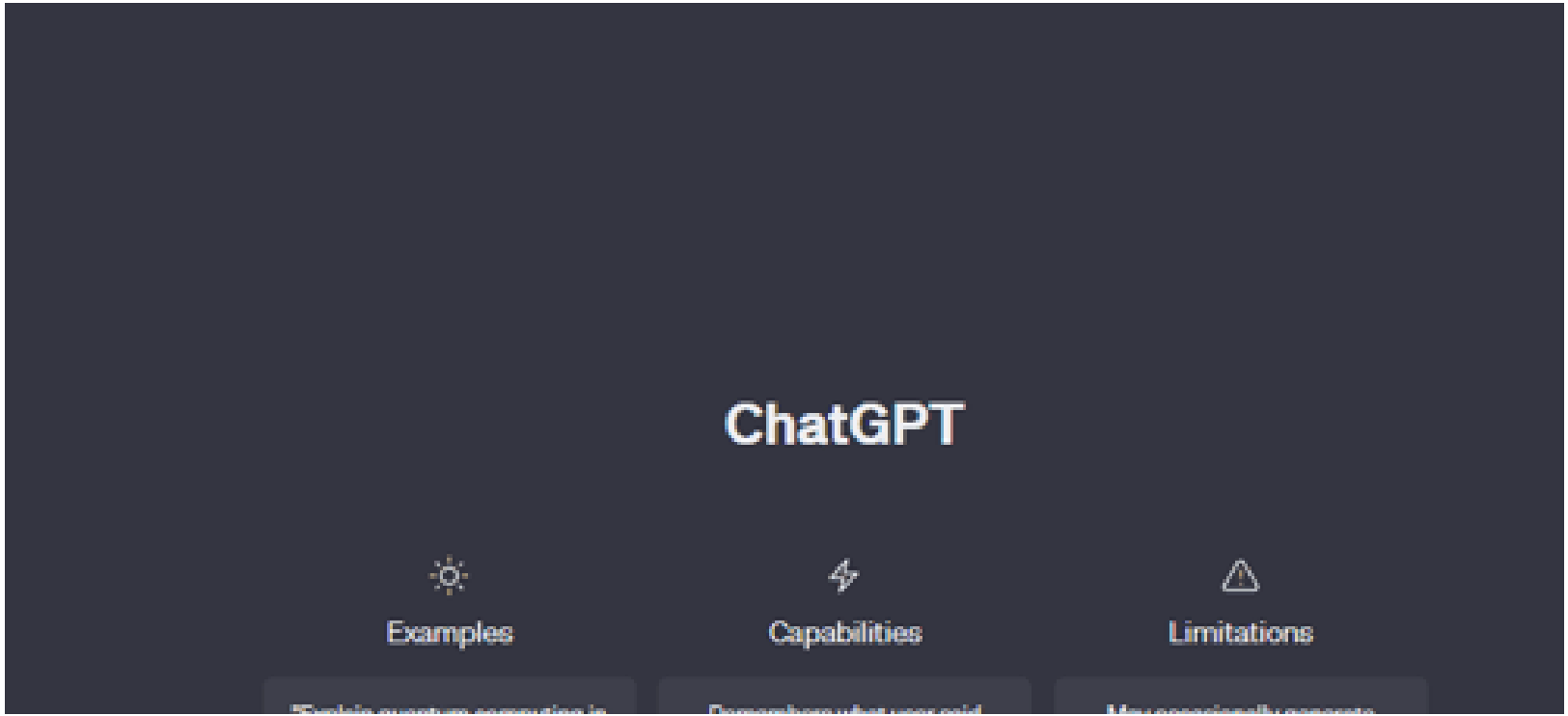
Visual Studio LIVE!
EXPERT SOLUTIONS FOR ENTERPRISE DEVELOPERS



Demo

Semantic Kernel Intro

Streaming content



Visual Studio LIVE!
EXPERT SOLUTIONS FOR ENTERPRISE DEVELOPERS

Tokens == cost $

My favorite color is red.

Text | Token IDs

[3666, 4004, 3124, 318, 2266, 13]

Text | Token IDs

https://platform.openai.com/tokenizer

Visual Studio LIVE!
EXPERT SOLUTIONS FOR ENTERPRISE DEVELOPERS

Adding Logging

```
var builder = Kernel.CreateBuilder();
    builder.Services.AddLogging(
        s => s.AddConsole().SetMinimumLevel(LogLevel.Trace));
    Kernel kernel =
        builder.AddOpenAIChatCompletion(deploymentName, apiKey).Build();
```

Configuration Temperature

The **temperature parameter** is a crucial dial in the engine of GPT models, controlling the randomness of predictions

Top P **defines a range of tokens (words and symbols) that ChatGPT can use**. When Top P = 1, the language model can use any token while generating a response. When Top P = 0.5, it can only use 50% of the most probable/appropriate/common options.

Use Case	Temperature	Top_p	Description
Code Generation	0.2	0.1	Generates code that adheres to established patterns and conventions. Output is more deterministic and focused. Useful for generating syntactically correct code.
Creative Writing	0.7	0.8	Generates creative and diverse text for storytelling. Output is more exploratory and less constrained by patterns.
Chatbot Responses	0.5	0.5	Generates conversational responses that balance coherence and diversity. Output is more natural and engaging.
Code Comment Generation	0.3	0.2	Generates code comments that are more likely to be concise and relevant. Output is more deterministic and adheres to conventions.
Data Analysis Scripting	0.2	0.1	Generates data analysis scripts that are more likely to be correct and efficient. Output is more deterministic and focused.
Exploratory Code Writing	0.6	0.7	Generates code that explores alternative solutions and creative approaches. Output is less constrained by established patterns.

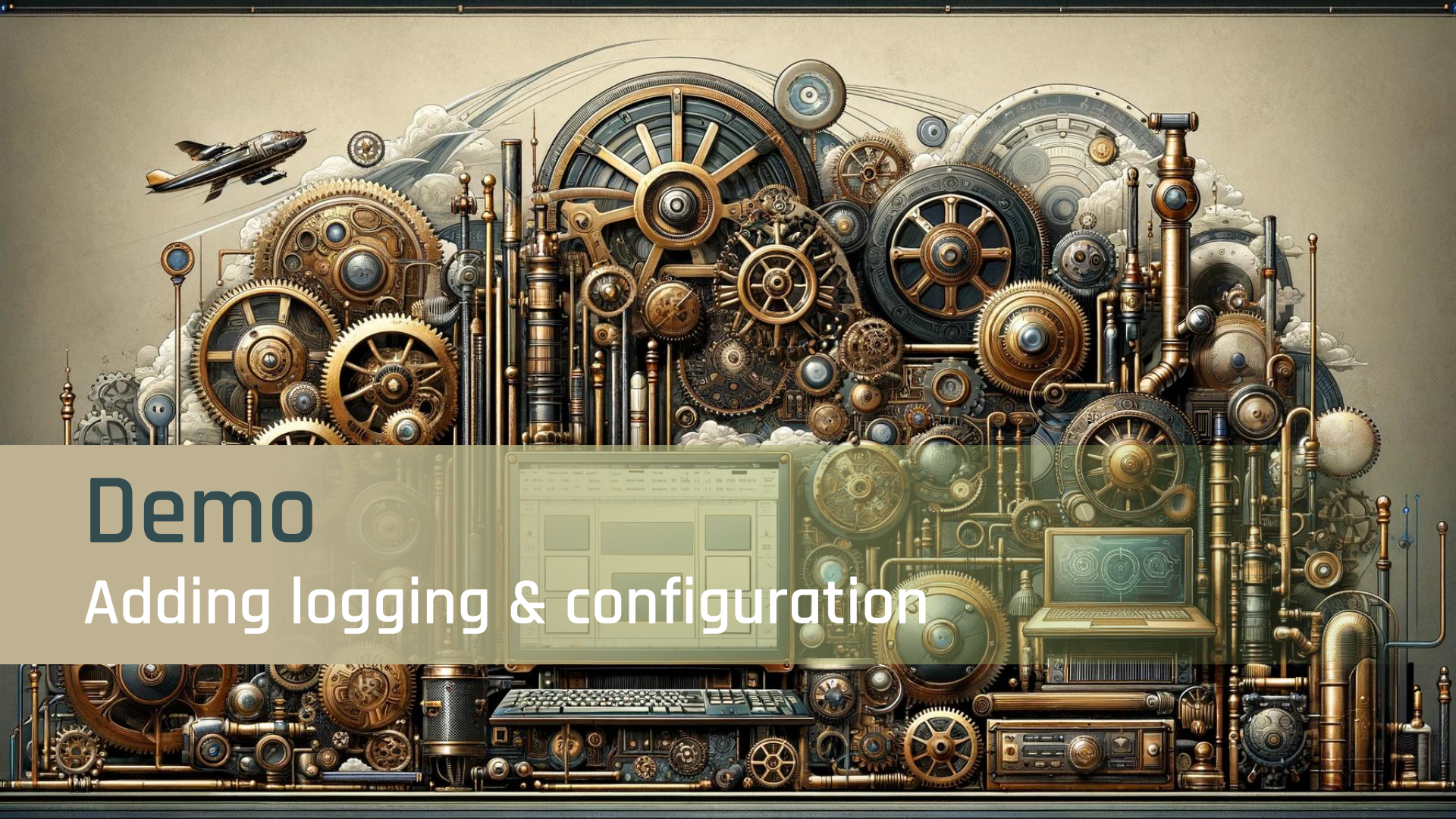
https://community.openai.com/t/cheat-sheet-mastering-temperature-and-top-p-in-chatgpt-api/172683

Adding Configuration

```
string response = string.Empty;
Kernel kernel = Kernel.CreateBuilder().
       AddAzureOpenAIChatCompletion(deploymentName, endpoint, apiKey).Build();
var chatCompletionService = kernel.GetRequiredService<IChatCompletionService>();

KernelArguments arguments = new(new OpenAIPromptExecutionSettings
{ MaxTokens = 500, Temperature = 0 });

Console.WriteLine(
       await kernel.InvokePromptAsync("Tell me about Orlando", arguments));
```



Demo

Adding logging & configuration

Adding more power

- The LLM only knows what it was trained on
- It can reason if you give it more information
- It can call functions if it knows how to get that information

Visual Studio LIVE!
EXPERT SOLUTIONS FOR ENTERPRISE DEVELOPERS

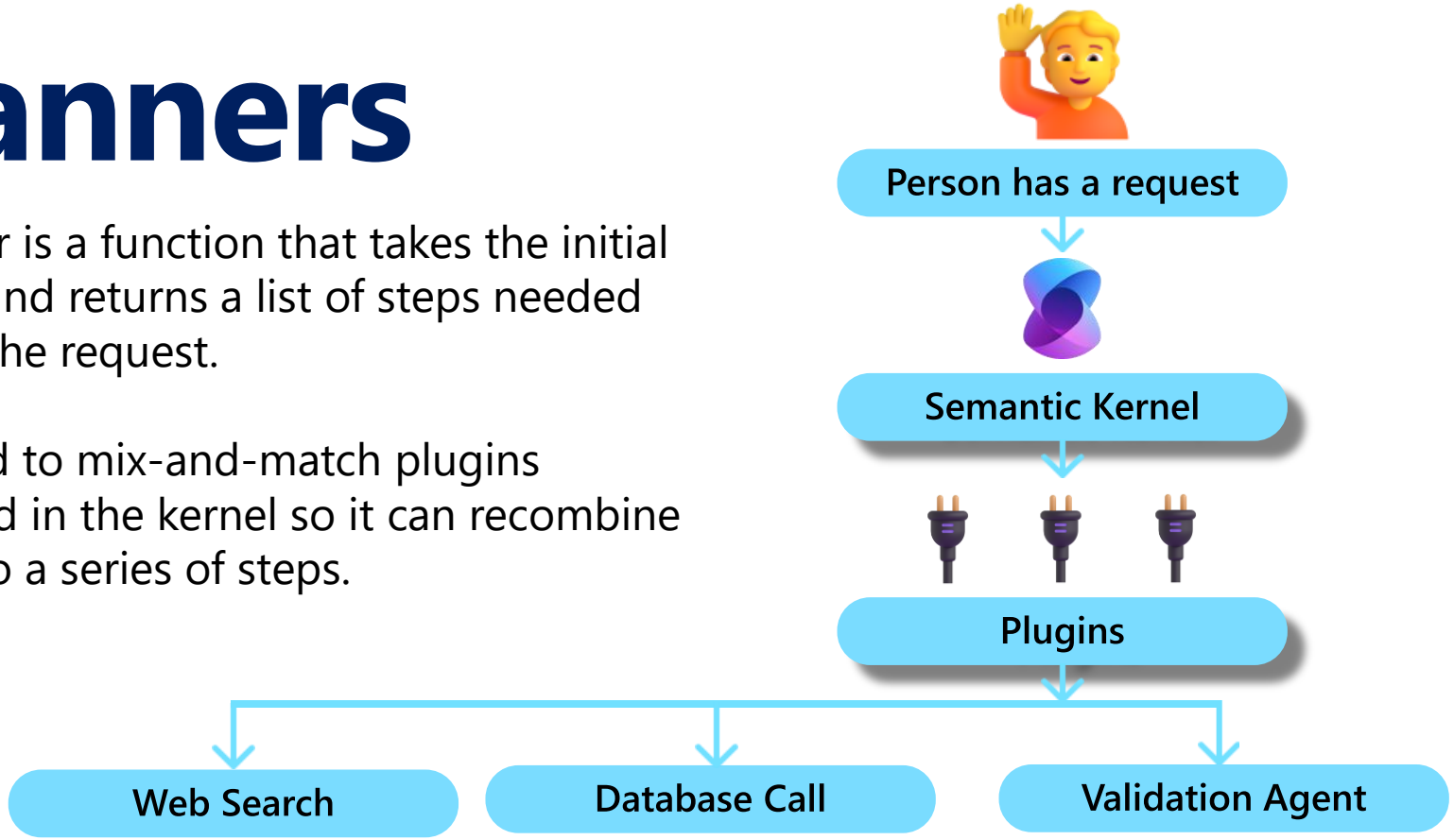
Knowledge Cutoff Dates

LLMs	Knowledge cutoff date	Provider
GPT-o1-preview	October 2023	OpenAI
GPT-4o	October 2023	OpenAI
GPT-4	April 2023	OpenAI
GPT-3.5	January 2022	OpenAI
Google Gemini Pro	April 2023	Google
Google PaLM 2	September 2022	Google
Llama 3 – 70B	December 2023	Meta
Claude 3	August 2023	Anthropic
Mistral – 7B	August 2021	Mistral

Planners

A Planner is a function that takes the initial request and returns a list of steps needed to fulfill the request.

AI is used to mix-and-match plugins registered in the kernel so it can recombine them into a series of steps.



Prompt to function calls

"What is the weather like today in Amsterdam?"

Person has a request

Semantic Kernel

Time Plugin

Weather Plugin

Visual Studio LIVE!
EXPERT SOLUTIONS FOR ENTERPRISE DEVELOPERS

Build your own plugins

```
public class TimePlugin
{
  [KernelFunction]
  [Description("Gets the current date and time in UTC")]
  public string GetCurrentDateAndTime()
  {
      return DateTime.UtcNow.ToString("R");
  }
}
```

Build your own plugins

```
public class WeatherPlugin
{
  [KernelFunction]
  [Description("Gets the weather on a given date and location")]
  public string GetWeather(
          [Description("The date to give the weather for")] string date,
          [Description("The location to give the weather for")] string location)
  {
    if (location == "Amsterdam")
    //it's always good weather to grow crops in Amsterdam...??? :-D
    return $"The weather in {location} on {date} is 25℃ sunny with a drop of rain";
    else
    {
      return "I'm sorry, I don't have that information.";
    }
  }
}
```

Visual Studio LIVE!
EXPERT SOLUTIONS FOR ENTERPRISE DEVELOPERS

Prompt to function calls

"What is the weather like today in Amsterdam?"

Person has a request

Semantic Kernel

Time Plugin

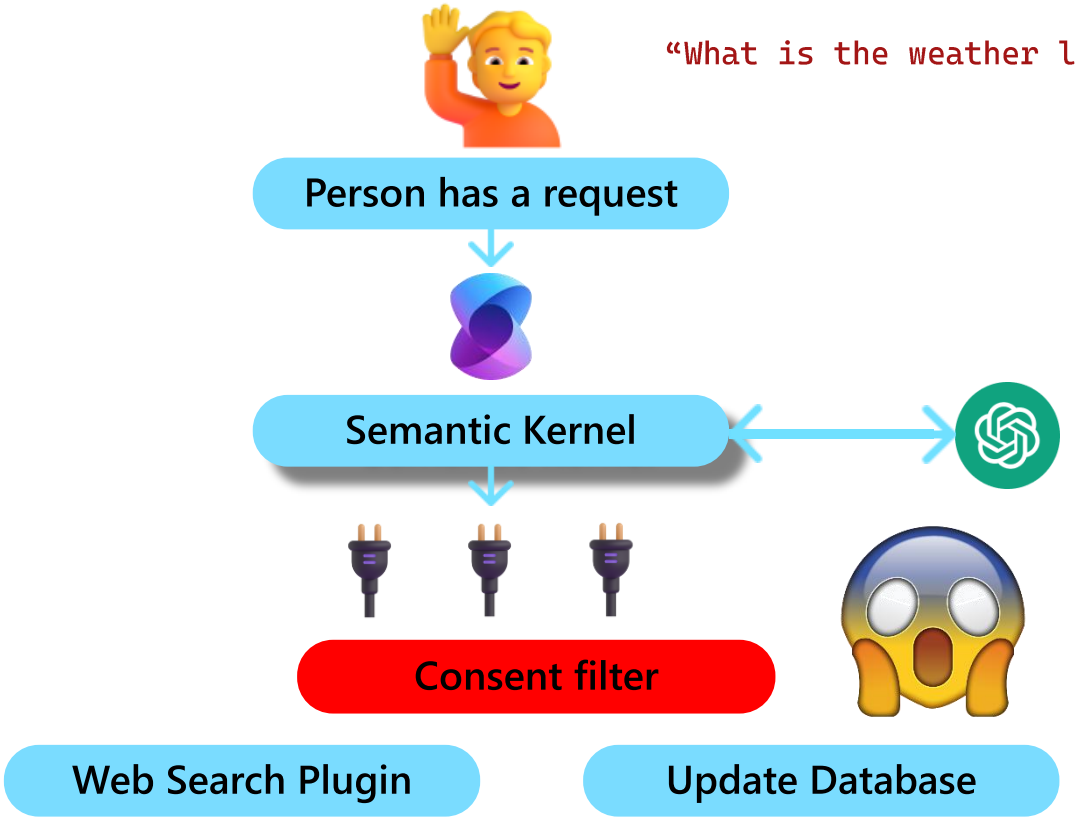
Weather Plugin

Visual Studio LIVE!
EXPERT SOLUTIONS FOR ENTERPRISE DEVELOPERS

Demo

Function calling

Can it call anything?



Visual Studio LIVE!
EXPERT SOLUTIONS FOR ENTERPRISE DEVELOPERS

User Consent

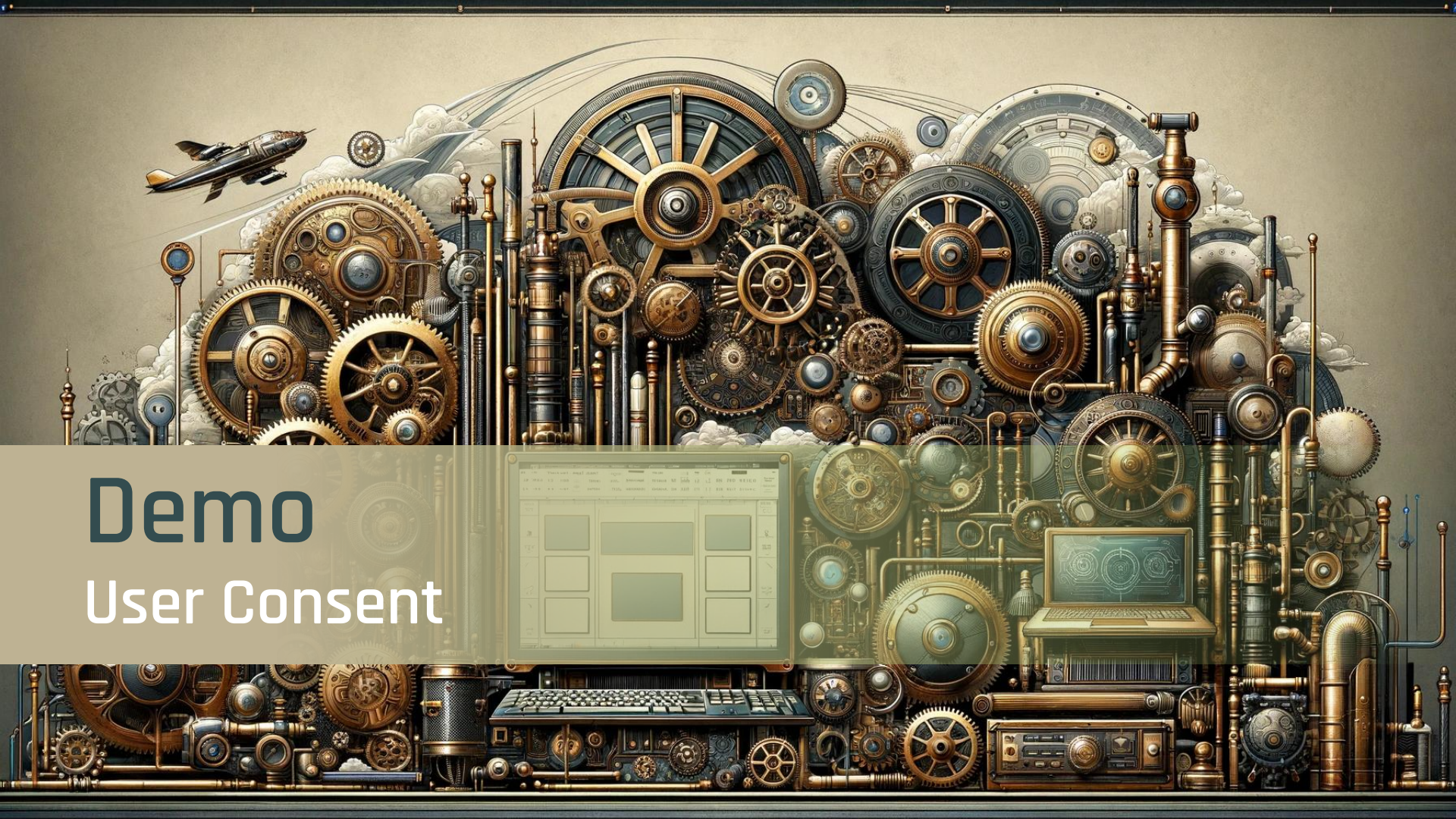
```
public class ApprovalFilterExample() : IFunctionInvocationFilter
{
  public async Task OnFunctionInvocationAsync(FunctionInvocationContext context,
                        Func<FunctionInvocationContext, Task> next)
  {
    Console.WriteLine($"System > The agent wants to call {context.Function.PluginName}
                              with function call {context.Function.Name}. Approve? (Y/N)");
    string shouldProceed = Console.ReadLine()!;
    if (shouldProceed.ToLower() != "y")
    {
        context.Result = new FunctionResult(context.Result, "Call Not allowed!");
        return;
    }
    await next(context);
  }
}
```

Visual Studio LIVE!
EXPERT SOLUTIONS FOR ENTERPRISE DEVELOPERS

Built-in plugins

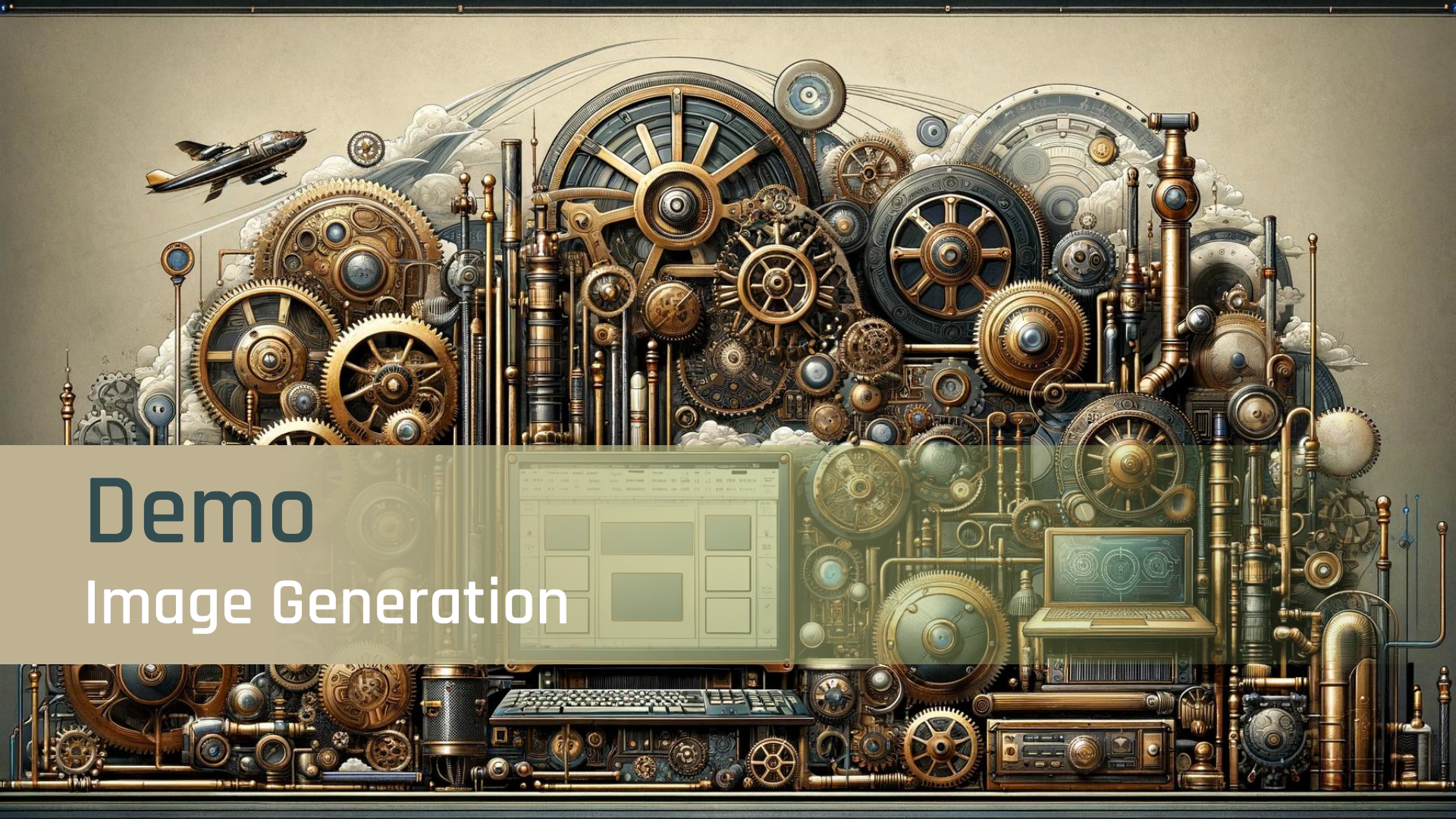
The Semantic Kernel SDK offers an extra package with predefined plugins for common tasks. These are available in the `Plugins.Core` package that you can install with NuGet: `dotnet add package Microsoft.SemanticKernel.Plugins.Core --version 1.2.0-alpha` The package includes the following plugins:

- `ConversationSummaryPlugin` - Summarizes conversation
- `FileIOPlugin` - Reads and writes to the filesystem
- `HttpPlugin` - Makes requests to HTTP endpoints
- `MathPlugin` - Performs mathematical operations
- `TextPlugin` - Performs text manipulation
- `TimePlugin` - Gets time and date information
- `WaitPlugin` - Pauses execution for a specified amount of time



Demo

User Consent

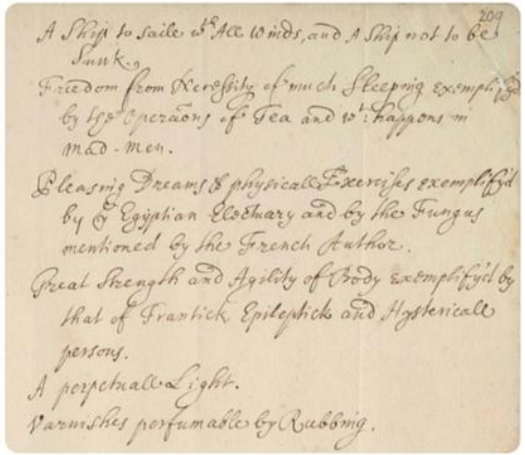


Demo

Image Generation

Multi Modal

- Multiple types of input / output
- Text, video, audio



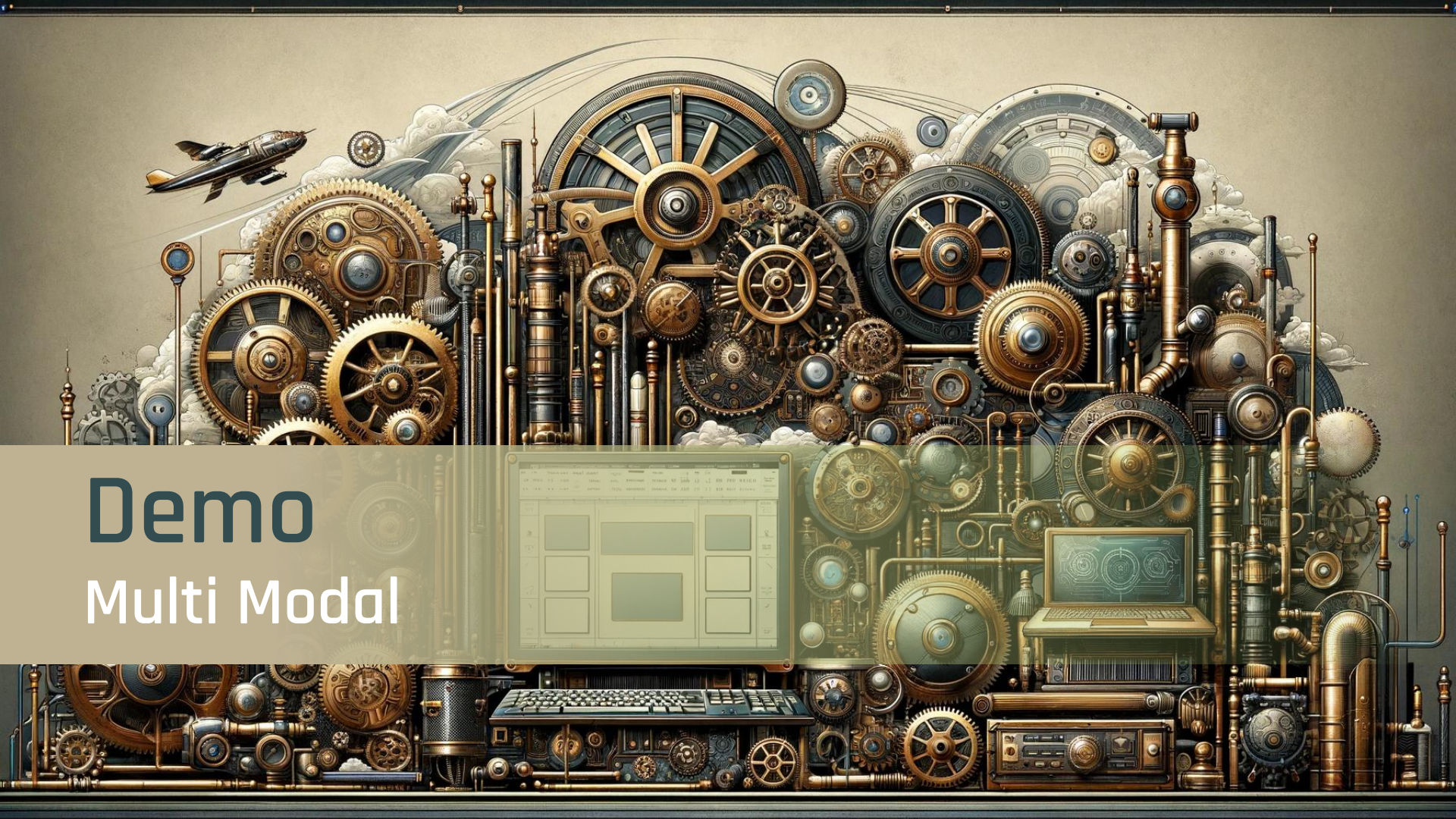
Can you read this?

CHATGPT

Yes, I can read the text in the image. Here's a transcription:

```
A Ship to saile without winds, and a Ship not
to be sunk.
Freedom from Necessity of much Sleeping
exemplified by the Operations of Tea and what
happens in Mad men.
Pleasing Dreams & physicall Exercises
exemplified by the Egyptian Electuary and by
the Fungus mentioned by the French Author.
Great Strength and Agility of Body
exemplified by that of Frantick Epileptick
and Hysterical persons.
A perpetuall Light.
Varnishes performable by Rubbing.
```

Visual Studio LIVE!
EXPERT SOLUTIONS FOR ENTERPRISE DEVELOPERS



Demo

Multi Modal

RAG

Retrieval Augmented Generation

I know things, I know how to reason about information I get
I was trained on C# programming and have a bachelors degree in electronics in 1996

What is the temperature in the room in Celsius?

I know to ask someone but get back 72 Fahrenheit

I know how to calculate Fahrenheit to Celsius

RAG

Retrieval Augmented Generation

I know things, I know how to reason about information I get
I was trained on C# programming and have a bachelors degree in electronics in 1996

What is the temperature in the room in Celsius?

Given the fact the last room temperature reading is 72 Fahrenheit

I know how to calculate Fahrenheit to Celsius: 22

$$°\text{Celsius} = \frac{5}{9} * (°\text{Fahrenheit} - 32)$$

Visual Studio LIVE!
EXPERT SOLUTIONS FOR ENTERPRISE DEVELOPERS

RAG

Retrieval Augmented Generation

I know things, I know how to reason about information I get
I was trained on the Internet and graduated somewhere in 2023

What is the policy on refunds on a booked room here?

Get the info from documents that describe the policy

I know how to interpret the document and draw conclusions

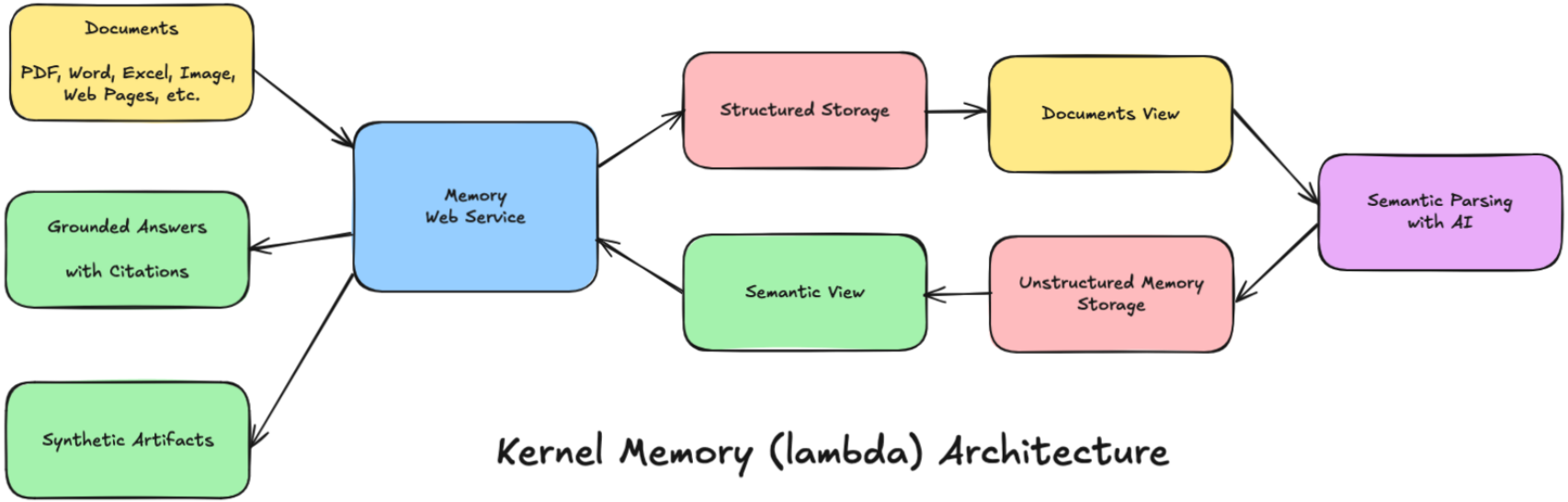
RAG

Retrieval Augmented Generation

- Creating Embeddings
- Semantic Search
- “Normal search”
- Vector Databases

RAG

Retrieval Augmented Generation

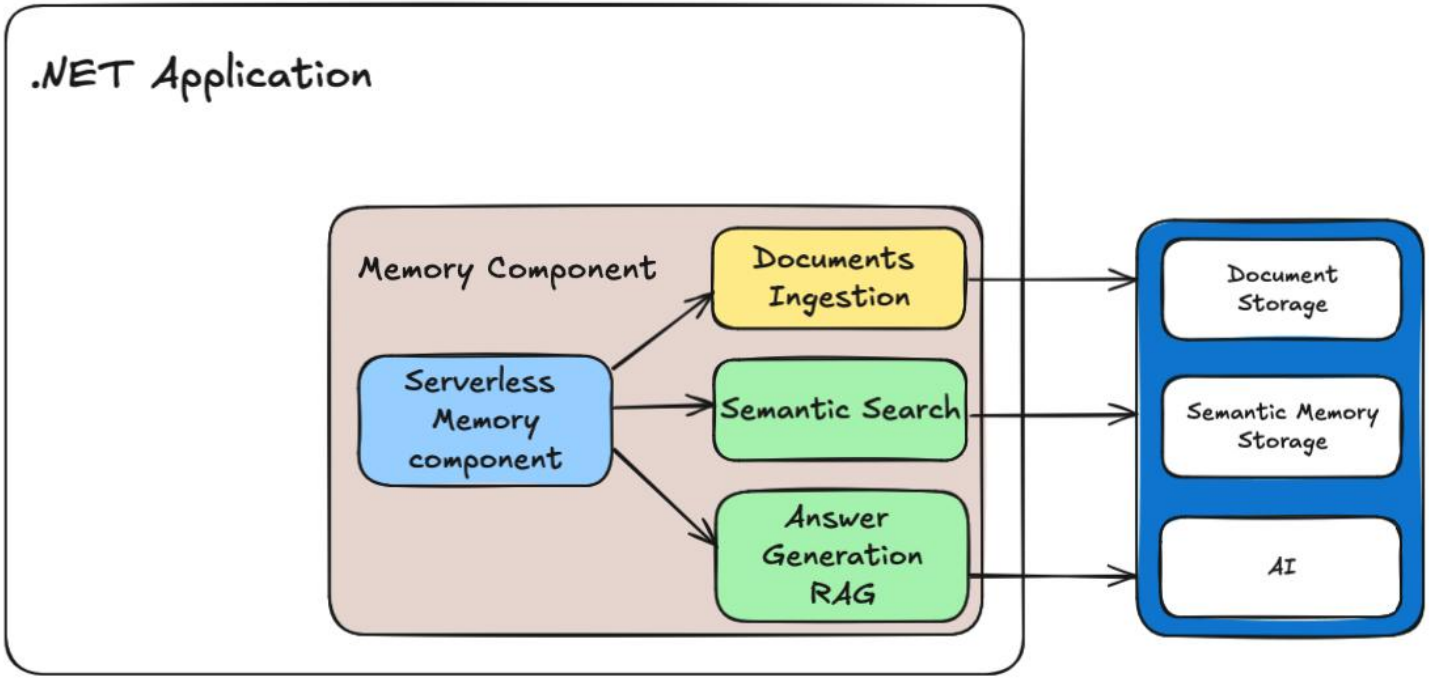


Kernel Memory Plugin

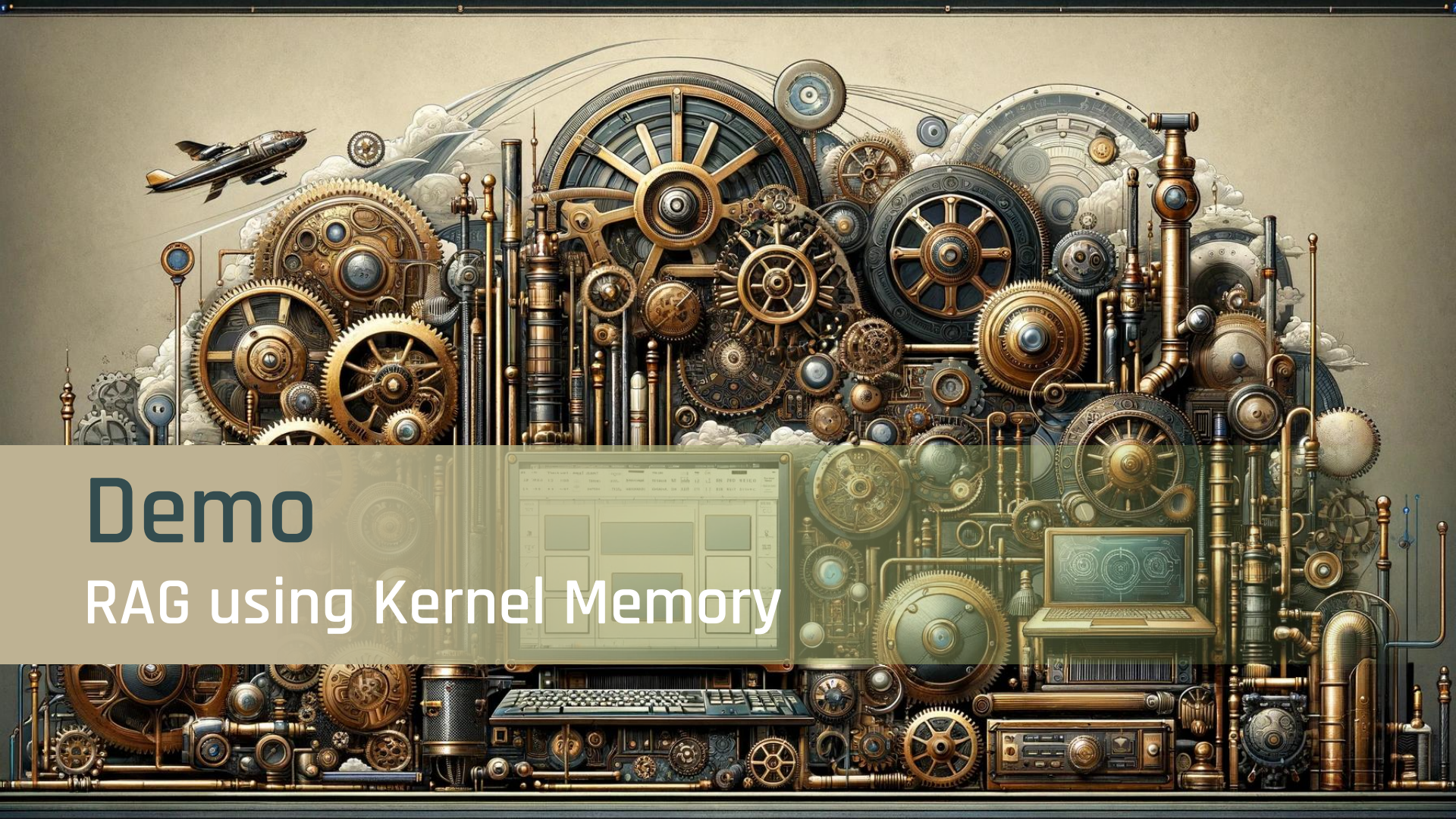
Visual Studio LIVE!
EXPERT SOLUTIONS FOR ENTERPRISE DEVELOPERS

Kernel Memory

Kernel Memory as an embedded .NET component

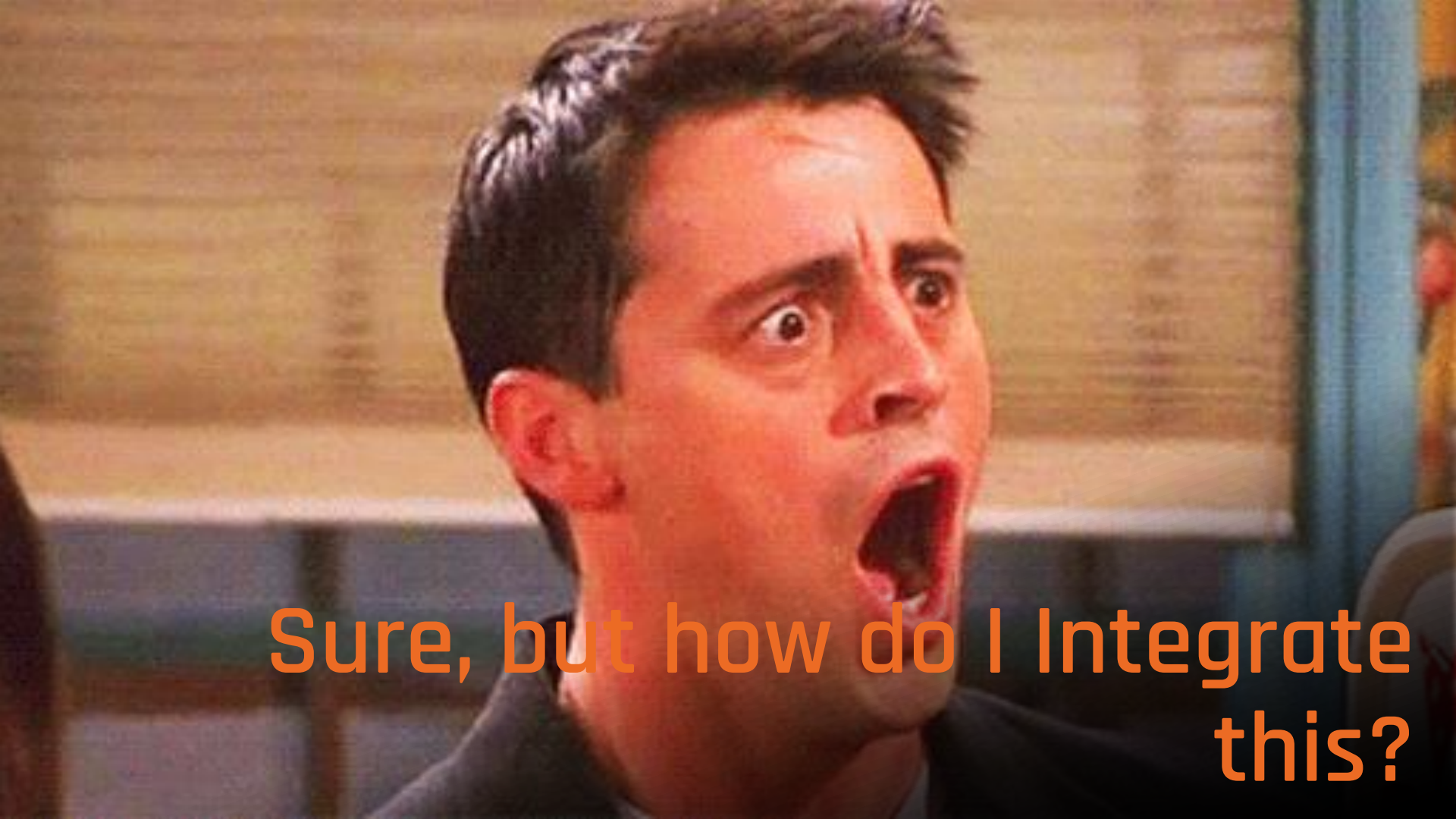


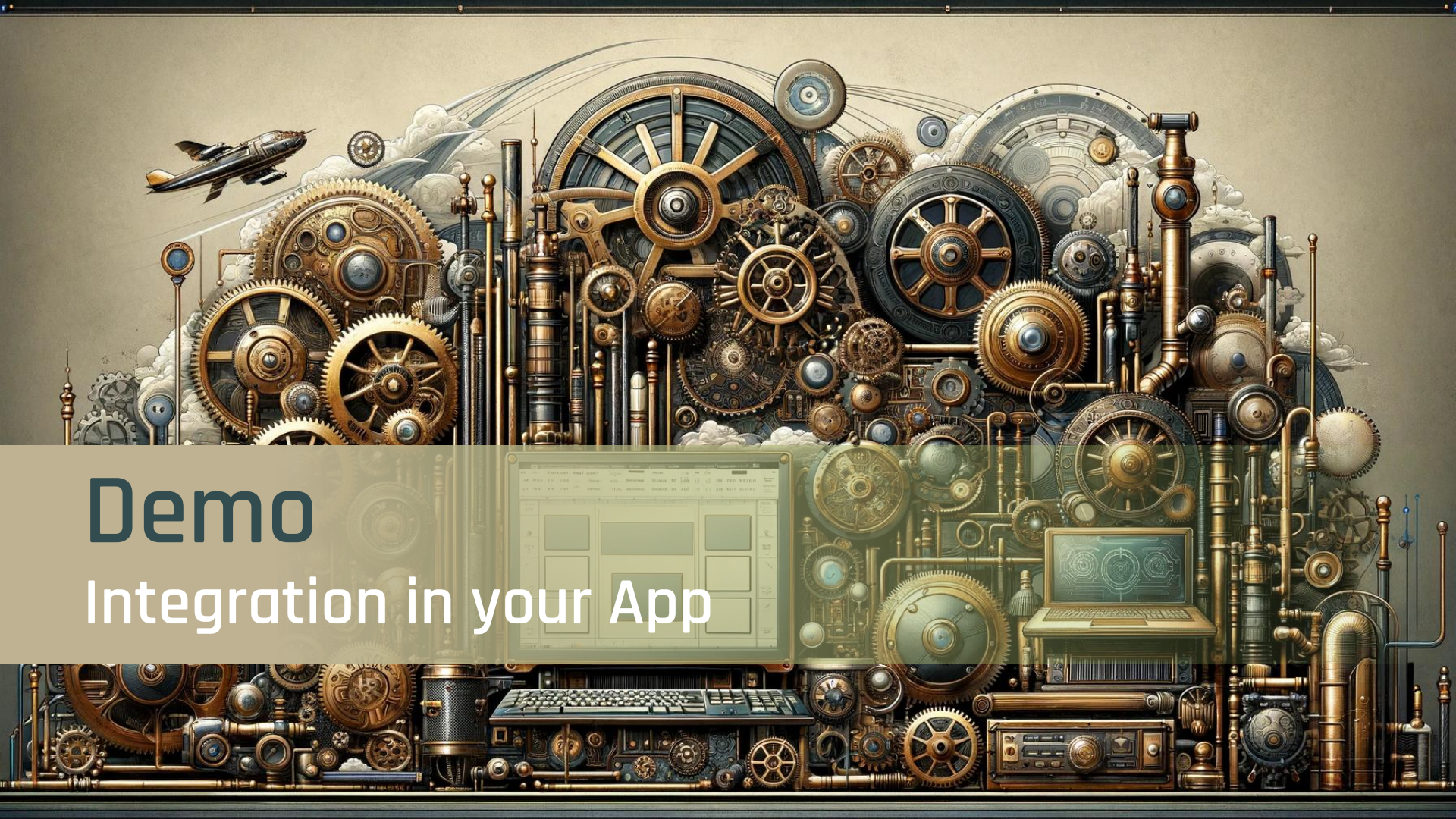
Visual Studio LIVE!
EXPERT SOLUTIONS FOR ENTERPRISE DEVELOPERS



Demo

RAG using Kernel Memory


Sure, but how do I Integrate this?



Demo

Integration in your App

Review my session

THANK YOU!

Marcel de Vries

Global MD & CTO Microsoft Services

@marcelv http://fluentbytes.com

Xebia | Microsoft Services

Microsoft REGIONAL DIRECTOR

MVP Microsoft® Most Valuable Professional

Attributions

unsplash.com, gratisography.com, mid-journey used for backgrounds

giphy.com for animated gifs

Where to find the code?

All demos:

- https://github.com/vriesmarcel/vslive-2026-vegas-sk